Ryan Dung Pham

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EDUCATION

California State Polytechnic University, Pomona — **Expected Graduation Date: December 2026**
B.S. Computer Science — GPA: 3.93
- Organizations: Google Developer Student Clubs, Computer Science Society, Data Science & AI Club
- Honors: President's List (2x), Dean's List (4x)

SKILLS

Languages: Python, Java, C++, C#, JavaScript, TypeScript, HTML, CSS, SQL
Technologies: React.js, Node.js, Express.js, Git, MongoDB, PostgreSQL, SQLite3, Redis, Selenium, Spring Boot

EXPERIENCE

CADENCE-SWANS CubeSat (AFRL University Nanosatellite Program) — **Pomona, CA**
Flight Software Engineer — May 2026 - Present
- Developed embedded flight software for a 12U CubeSat pathfinder spacecraft using NASA JPL's F Prime (F') framework, designing modular C++ components, ports, and system topologies for mission-critical operations.
- Defined component interfaces, ports, commands, and telemetry channels to support scalable flight software integration and system-level verification.
- Designed and executed unit and integration tests to verify software functionality, interface behavior, and subsystem interactions throughout the development lifecycle.
- Collaborated with multidisciplinary teams across flight software, avionics, and systems engineering to support mission development under the University Nanosatellite Program (UNP).
- Presented software architecture decisions, development progress, and verification results during technical reviews with AFRL stakeholders and engineering leadership.

CIC | PCUBED Data Science Summer Research Project — **Fullerton, CA**
Research Scholar — June 2023 - August 2023
- Engineered predictive models in scikit-learn, achieving 80% accuracy in forecasting New York City shooting incidents after implementing feature engineering on 28K+ police records (temporal, spatial, and demographic variables).
- Optimized data integrity by cleansing and preprocessing inconsistent values, improving training efficiency and ensuring reproducibility of results.

Code Ninjas — **Fountain Valley, CA**
Programming Instructor — March 2023 - April 2024
- Designed and delivered comprehensive curriculum covering Python, JavaScript, Java, and C# for 100+ students ranging from pre-K to high school level.
- Led specialized development camps in Unity (C#), TinkerCad, and Roblox Studio (Lua), achieving 95% student retention rate through interactive teaching methodologies.

NASA L'SPACE Academy — **Remote**
Research Scholar — June 2023 - August 2023
- Contributed to 13-person interdisciplinary team developing Mars mission concept for crustal magnetic field analysis in Southern Highlands region, focusing on programmatics and computer hardware systems.
- Conducted cost analysis and technical feasibility assessment for $300M Discovery-class mission, supporting payload design and dual-rover deployment system for subsurface ice detection and atmospheric analysis.

PROJECTS

Mikomi (Python, TypeScript, React, Node.js, FastAPI, ArangoDB, Sentence-BERT):
- Developed full-stack AI-powered manga recommendation engine serving personalized content discovery for manga enthusiasts. Engineered machine learning pipeline using Sentence-BERT to embed manga synopses into high-dimensional vector spaces, implementing multi-layered similarity scoring (cosine, Jaccard, weighted Jaccard) for semantic content matching. Built scalable FastAPI backend with ArangoDB integration for efficient vector storage and retrieval. Designed responsive React frontend with TypeScript enabling real-time search functionality and dynamic recommendation filtering.

Buzzr (JavaScript, SQLite3, Node.js, Twilio API, Discord.js):
- Architected and deployed scalable Discord bot serving 100+ users across 2 sports community servers, increasing membership through enhanced notification capabilities.
- Engineered real-time SMS notification system using Twilio API, processing 10+ daily alerts and enabling instant mobile notifications for improved user engagement.

CineMatch (Python, FastAPI, React, OpenAI API, Model Context Protocol, TMDB API, SSE, AsyncIO, Pydantic, REST):
- Architected a full-stack agentic AI application across 3 services (React frontend, FastAPI backend, MCP tool server) using REST APIs, SSE, and OpenAI tool calling to deliver personalized movie/TV recommendations grounded in live external data.
- Built a Model Context Protocol (MCP) server with 3 schema-validated tools routing 100% of TMDB access through a single layer; required per-title detail/trailer tool calls so recommendations use verified metadata (posters, runtime, ratings) instead of LLM hallucination.
- Reduced TMDB search/discover latency by ~50% via parallel async HTTP (httpx, asyncio.gather) across movie and TV endpoints; implemented an agent loop supporting up to 24 concurrent tool-call rounds with structured JSON output.